Nithin Gowda B.S.

Kunigal, Tumkur, Karnataka 572130 • +91 9448443305 • nithingowdabs279@gmail.com

Research Focus: MOF-Based Electrochemical Biosensing | Cancer Biomarkers | Heavy Metal Detection | Point-of-Care Diagnostics

EDUCATION

CHRIST (Deemed to be University), Bangalore 2024 – 2027 (Expected)

Bachelor of Science in Chemistry & Zoology | GPA: 3.3 / 4.0

Undergraduate Researcher | Research & Development

Sarvodaya PU College, Tumkur 2024

Pre-University Certificate (12th Grade), Karnataka State Board | 89%

Arvind International School, Kunigal 2022

Secondary School Certificate (10th Grade), ICSE Board | 91%

RESEARCH EXPERIENCE

Lead Undergraduate Researcher — Electrochemical Cancer Biomarker Detection 2025 – Present

Dr. Priya Vijayaraghavan (DBT Ramalingaswami Re-entry Fellow) | CHRIST University | DBT-Funded Project (₹89 Lakhs)

- Developed Fe-Ni bimetallic MOF/conducting polymer composites on carbon cloth for electrochemical detection of cancer biomarkers
- Led the complete research pipeline independently: MOF synthesis, electrode fabrication, characterization (SEM, XRD, FTIR, UV-Vis, CV, EIS), and manuscript writing
- Manuscript in preparation for submission to a high-impact peer-reviewed journal

Self-Initiated Research — Hormone & Glucose Sensing 2025 – Present

Dr. Priya Vijayaraghavan | CHRIST University | Self-initiated with supervisor approval

- Extended the Fe-Ni MOF platform independently to electrochemical detection of AMH, dopamine, and glucose
- Demonstrated scientific initiative and independent research capability beyond the funded project scope

Self-Initiated Research — Heavy Metal & Rare Earth Metal Detection 2025 – Present

Dr. Priya Vijayaraghavan | CHRIST University | Self-initiated with supervisor approval

- Developing Fe-Ni, Co-Ni, and Co-Cu bimetallic MOF systems for simultaneous electrochemical detection of Pb, Cd, and rare earth metals
- Fabricated and characterised modified electrodes on carbon cloth, nickel foam, and copper foam substrates

Undergraduate Researcher — Biochar-Based Electrochemical Sensing 2025 (45 days)

Dr. Anitha Varghese (HOD, Stanford Top 2% Researcher) | CHRIST University

- Synthesised biochar from pineapple leaf waste and deposited onto carbon fiber paper electrodes for laccase detection
- Conducted material synthesis, electrode surface modification, cyclic voltammetry (CV), and EIS analysis

PUBLICATIONS

Under Review

1. Gowda N., Shaju E.M., Vijayaraghavan P., Cherian A.R. *"Ambient Electrochemical Nitrogen Reduction to Ammonia: Catalyst Design, Selectivity Challenges, and Mechanistic Insights."* (Under Review, 2025)

2. Gowda N., Udupa P., Vijayaraghavan P. *"CRISPR-Cas Integrated Electrochemical Biosensors: Mechanisms, Design Strategies, and Applications in Diagnostics, Environmental Monitoring, and Food Safety."* (Under Review, 2025)

CONFERENCES & PRESENTATIONS

SAINTS 2026 — Scientific Advances in Natural and Technological Sciences January 2026

CHRIST University, Bangalore | International Conference | Poster Presentation

AMEEBA-2026 — Advanced Materials for Energy, Environment and Biomedical Applications May 2026

JSS Academy of Technical Education, Bengaluru | National Conference | Poster Presentation

NCURB 2025 — National Conference for Undergraduate Research in Biosciences December 2025

Indian Institute of Science Education and Research (IISER), Tirupati | National Conference | Poster Presentation

AWARDS & HONORS

- 2nd Prize — Researcher's Day, CHRIST University (Poster: Monometallic MOF for Cancer Detection)
- 4th Place — National Science Day, CHRIST University (Poster: Fe-Ni MOF Biosensing Project)
- Selected for the GSI FAIR Research Program, GSI Helmholtz Centre for Heavy Ion Research, Darmstadt, Germany — one of the world's leading facilities for heavy ion physics and materials science research (unable to attend due to ongoing research commitments)

PROFESSIONAL MEMBERSHIPS

- Student Member, Royal Society of Chemistry (RSC) | December 2025

CERTIFICATIONS & ADDITIONAL COURSES

- Mental Health and Wellbeing — IIT Kanpur, NPTEL
- Basic Python Programming — CHRIST University Curriculum

TECHNICAL SKILLS

Synthesis: MOF synthesis (Fe-Ni, Co-Ni, Co-Cu bimetallic), biochar synthesis, conducting polymer composites

Electrode Fabrication: Carbon cloth, screen-printed carbon electrodes (SPCE), carbon fiber paper, nickel foam, copper foam

Electrochemical Methods: Cyclic Voltammetry (CV), Electrochemical Impedance Spectroscopy (EIS)

Characterization: Scanning Electron Microscopy (SEM), X-Ray Diffraction (XRD), FTIR, UV-Vis Spectroscopy

Software: Python (basic), Origin Pro (data analysis)

LANGUAGES

English (Fluent) | Kannada (Native) | Hindi (Basic) | Telugu (Basic) | French (Basic)

REFERENCES & LETTERS OF RECOMMENDATION

- Dr. Priya Vijayaraghavan — DBT Ramalingaswami Re-entry Fellow, CHRIST University
- Dr. Anitha Varghese — Head of Department, Sciences, CHRIST University (Stanford Top 2% Researcher)

- Dr. Vinod T.P. — CHRIST University (Stanford Top 2% Scientist)